

EEG pattern recognition based on Dual-Tree Complex Wavelet Transform and Particle Swarm Optimization

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Abstract—Aiming at the issue of motor imagery electroencephalography (EEG) pattern recognition in the research of brain-computer interface (BCI), a novel method based on dual-tree complex wavelet transform (DTCWT) and particle swarm optimization (PSO) was proposed. The advantage of DTCWT over discrete wavelet transform (DWT) was discussed in depth and the ERD/ERS phenomenon was verified at first. Then, the signal component related to sensory motor rhythms was extracted based on dual tree complex wavelet decomposition and reconstruction. Afterward, PSO algorithm was implemented to search the optimal time interval automatically for feature extraction. Finally, average energy, root mean square and signal variance were extracted as features and linear discriminant analysis (LDA) was applied for classification. The results show that the proposed method can find the relatively optimal time interval for feature extraction automatically and the maximum classification accuracy is 90%, which is better than the BCI competition winner.

Keywords—brain computer interface; motor imagery; dual-tree complex wavelet transform; particle swarm optimization; electroencephalography

I. INTRODUCTION

Brain-computer interfaces (BCIs) use electroencephalography (EEG) or other electrophysiological measures of brain activity to establish a non-muscular channel for sending instructions or commands to external devices [1]. The scalp-recorded EEG from a participant trying to convey his motor intentions by motor imagery is a most widely used signal for noninvasive BCI control. Motor imagery (MI) based BCI uses the information correlated with amplitude modulations of sensory motor rhythms (SMR) to reflect the motor intention of a subject [2]. The modulation of SMR generates contra lateral preponderant event related desynchronization (ERD) and synchronization (ERS) which are short lasting amplitude attenuation and enhancements in the rhythmic components of EEG [3]. Currently, ERD/ERS based BCIs have received a lot of attentions due to their potential application in motor rehabilitation and their assisting for the motor function impaired patients [4].

Generally, a BCI system contains signal pre-processing, feature extraction and classification. Due to the nonstationarity

and low signal-to-noise ratio of EEG signals, an effective feature extraction algorithm has great impact on the performance of the BCI system. A great variety of features have been used to design BCI such as amplitude values of EEG signals, band powers (BP), power spectral density (PSD) values, auto regressive (AR), adaptive auto regressive (AAR) and time-frequency features. The discrete wavelet transform (DWT) is a common feature extraction method, which is quite powerful in selecting features from the time-frequency domains, and DWT is an efficient and structured approach for ERD/ERS representation [5]. For the EEG signal classification issue, Linear Discriminant Analysis (LDA), Support Vector Machine (SVM) and Neural Networks (NN) are widely used. Apart from classification accuracy, design complexity and classification speed are also vital for classification algorithms.

Considering that traditional DWT is not able to cancel the aliasing, thus resulting in unclearly separated sub-bands. In this paper, dual-tree complex wavelet transform (DTCWT) [6] which was first introduced by Kingsbury, is proposed to extract the signal component related to sensory motor rhythms. It overcomes the mentioned shortcoming of DWT. In addition, to extract effective features and reduce computation burden, the optimal temporal segment for MI classification is always determined before feature extraction. In this paper, we treat this issue as an optimization problem and propose a particle swarm optimization (PSO) [7] based method to search the global optimum. Experimental results demonstrated that our proposed method can find the relatively optimal time interval for feature extraction automatically and provided high classification accuracy, which was better than the BCI competition winner.

II. MATERIALS AND METHODS

A. Data Description

To evaluate the effectiveness of our method, data set III of BCI Competition 2003 was used. In this data set, EEG signals were recorded over C3, Cz and C4 from a 25 years old healthy woman during a feedback session. The task was to control a feedback bar by means of imagery left or right hand movements. The experiment consisted of 7 runs with 40 trials each, so there were totally 280 trials of 9s length. The first 2s

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was quite, at $t=2s$ an acoustic stimulus indicated the beginning of the trial, and a cross was displayed for 1s, then at $t=3s$ an arrow (left or right) was displayed as cue which instructed the subject to perform left or right hand MI. The recording was made using a G.tec amplifier and Ag/AgCl electrodes. The EEG was sampled with 128Hz and filtered between 0.5 and 30Hz. Note that EEG signal in Cz electrode was non-related for left/right hand classification, in this study EEGs in only C3 and C4 electrodes were analyzed. In this competition, 140 trials were randomly selected as training set, and the rest 140 trials were for testing, the true labels of testing set were announced after the competition.

B. Dual-Tree Complex Wavelet Transform

DTCWT is an enhancement of the DWT. It employs two real DWTs, which give the real part and imaginary part of the transform, respectively. The dual-tree complex wavelets have the expression as below

$$\varphi(t) = \varphi_h(t) + i\varphi_g(t) \quad (1)$$

Where $\varphi_h(t)$ and $\varphi_g(t)$ denote the two real wavelets, respectively. Then the wavelet coefficients and scaling coefficients of real and imaginary parts can be obtained as in

$$d_j^{\text{Re}}(n) = 2^{j/2} \int_{-\infty}^{+\infty} x(t)\varphi_h(2^j t - n)dt, j = 1, 2, \dots, J \quad (2)$$

$$c_j^{\text{Re}}(n) = 2^{J/2} \int_{-\infty}^{+\infty} x(t)\phi_h(2^J t - n)dt \quad (3)$$

$$d_j^{\text{Im}}(n) = 2^{j/2} \int_{-\infty}^{+\infty} x(t)\varphi_g(2^j t - n)dt, j = 1, 2, \dots, J \quad (4)$$

$$c_j^{\text{Im}}(n) = 2^{J/2} \int_{-\infty}^{+\infty} x(t)\phi_g(2^J t - n)dt \quad (5)$$

Where ϕ_h and ϕ_g are corresponding scaling functions. Afterward, the wavelet coefficients and scaling coefficients of the dual-tree complex wavelets can be expressed as in

$$d_j^{(\varphi)}(n) = d_j^{\text{Re}}(n) + id_j^{\text{Im}}(n) \quad (6)$$

$$c_j^{(\phi)}(n) = c_j^{\text{Re}}(n) + ic_j^{\text{Im}}(n) \quad (7)$$

Finally, the wavelet coefficients and scaling coefficients above can be obtained via the inner products

$$d_j(t) = 2^{(j-1)/2} \left[\sum_{n=-\infty}^{\infty} d_j^{\text{Re}}(n)\varphi_h(2^j t - n) + \sum_{k=-\infty}^{\infty} d_j^{\text{Im}}(n)\varphi_g(2^j t - k) \right] \quad (8)$$

$$c_j(t) = 2^{(J-1)/2} \left[\sum_{n=-\infty}^{\infty} c_j^{\text{Re}}(n)\phi_h(2^J t - n) + \sum_{k=-\infty}^{\infty} c_j^{\text{Im}}(n)\phi_g(2^J t - k) \right] \quad (9)$$

Signal can be reconstructed by

$$x(t) = d_j(t) + c_j(t) \quad (10)$$

Note that we can obtain signal in each sub-band easily by setting the wavelet coefficients of other sub-bands to zero.

C. Sensory Motor Rhythms Extraction Based on DTCWT

MI is a common mental task that is being widely used among BCI research groups. The underlying basis of using hand MI as a BCI strategy is that it induces contra preponderant ERD and ERS which are short lasting amplitude attenuation and enhancements in the rhythmic components of EEG. ERD and ERS are two physiological phenomena that are closely related to two sensory motor rhythms, i.e. mu and beta rhythms. Generally, mu and beta rhythms are located in the frequency ranges of 8-12Hz and 14-35Hz, respectively.

To extract the sensory motor rhythms, EEG signal is decomposed into 4 levels. The frequency ranges represented by the wavelet coefficients A_4, D_4, D_3, D_2, D_1 are 0-4Hz, 4-8Hz, 8-16Hz, 16-32Hz and 32-64Hz, respectively. Since 8-16Hz contains mu rhythm and partial beta rhythm, signal is reconstructed based on D_3 for further analysis. To verify that DTCWT performs better than DWT in aliasing canceling, both methods are used, a q-shift method is taken to design the dual-tree filters and db5 wavelet is chosen as the wavelet basis function for DWT. Fig. 1 shows the original EEG signal and its D_3 reconstructed signals by DWT and DTCWT. The time domain waveforms of two reconstructed signals are quite similar, then we analyze them in frequency domain. In order to make comparison, EEG signal is also filtered between 8-16Hz using a fifth order butterworth band pass filter. The frequency spectrums of DWT reconstructed signal, DTCWT reconstructed signal and band pass filtered signal are shown in Fig. 2.

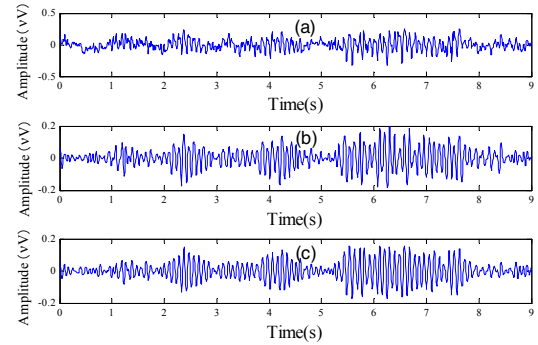


Fig. 1. Time domain waveforms of (a)original EEG signal, D_3 reconstructed signals by (b)DWT and (c)DTCWT.

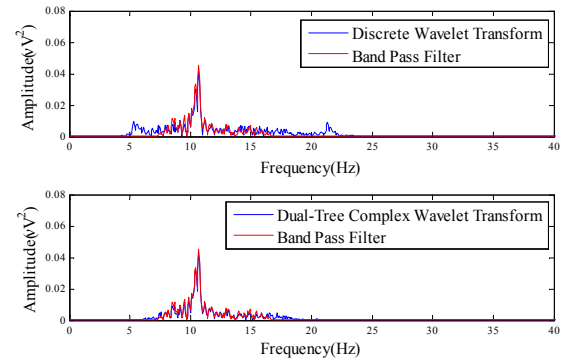


Fig. 2. Comparison of frequency spectrums of DWT reconstructed signal, DTCWT reconstructed signal and band pass filtered signal.

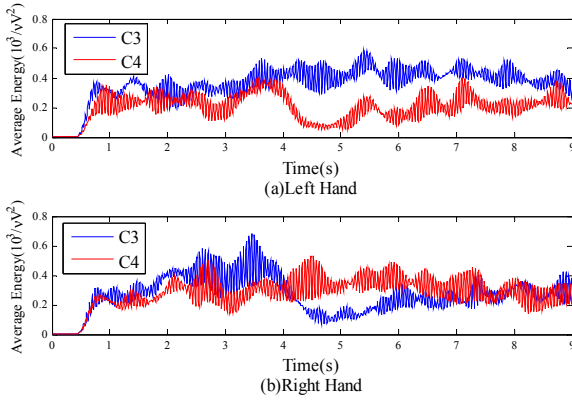


Fig. 3. Event related desynchronization (ERD) and synchronization (ERS) phenomena

From Fig. 2 we can see that due to the wide spacing of the wavelet coefficient samples, or equivalently, the fact that the wavelet coefficients are computed via iterated discrete-time down-sampling operations interspersed with nonideal low-pass and high-pass filters, substantial aliasing and artifacts are involved in the DWT reconstructed signal. It can be seen from the frequency spectrum that spurious frequency components, i.e., 5Hz and 22Hz are introduced in DWT reconstructed signal, while there are little frequency spectrum aliasing and leakage problems in DTCWT reconstructed signal. In DTCWT, the influence of wide spacing of the wavelet coefficient samples is reduced by combining the information of real and imaginary parts. Moreover, clearly separated sub-bands can be obtained via DTCWT, which is of great significance for subsequent feature extraction.

To further verify ERD/ERS phenomenon, for each trial in the training set, we first extract 8-16Hz signal component of C3 and C4 using DTCWT. These signals are then grouped according to MI classes (70 right hand MI trials, 70 left hand MI trials). Finally, average energy of each class is calculated. Fig. 3 shows the energy change of each class. From Fig. 3 we can see that, when people perform left hand motor imagery, the energy of C3 which is over the ipsilateral side increases while the energy of C4 which is over the contralateral side decreases, and the similar phenomenon can be observed when right hand motor imagery is performed.

D. Optimal Time Interval Searching Based on PSO

From Fig. 3 we can see that ERD/ERS phenomenon is quite notable in certain period and EEG signal in this time interval provides the most useful information for classification. In essence, the ultimate goal for searching optimal time interval is convergent to the optimization problem that how to find the global optimal time interval for feature extraction and classification. PSO is an evolution optimization algorithm that is not necessary to know the expression between object function and concerned variables. PSO has been proven to be powerful to search for the global optima. Based on the training set, we will use PSO to find the optimal time interval $[t_{strat}, t_{start} + t_{length}]$ for classification, where t_{strat} is the low

bound of time interval and t_{length} is the length of time interval. To evaluate a certain time interval $[t_{strat}, t_{start} + t_{length}]$, average classification accuracy of 10×5-fold cross validation will be served as fitness function. Obviously, the dimension of particle in PSO is 2, with each component representing t_{strat} and t_{length} . Considering that the subject is instructed to perform MI after $t=3s$, the searching ranges of t_{strat} and t_{length} are set to $[3 \times 128, 7 \times 128]$ and $[0 \times 128, 6 \times 128]$, respectively. The range of the velocity of the particle is set to $[-1 \times 128, 1 \times 128]$. The other parameters in PSO are initialized as: the swarm size Q is 250, the velocity constants C_1 and C_2 are both 2.0, the start point of inertia weight w_{start} is 0.9, the stop point of inertia weight w_{end} is 0.4, the maximum iteration number $MaxIter$ is 15. The PSO based approach for searching optimal time interval can be denoted as below.

Step 1. Particle initialization: Initialize each particle with 2 random values within their searching ranges. If $t_{strat} + t_{length} > 128 \times 9$, $t_{length} = 128 \times 9 - t_{strat}$. Then the quality of each particle is evaluated by calculating its fitness value.

Step 2. Best position recording: For all particles in the swarm after initialization, search the particle with the best fitness value, and this particle is reserved as the global optimum $gbest$.

Step 3. Update each particle: Update the inertia weight as,

$$w_{now} = ((w_{start} - w_{end}) * (MaxIter - Iter)) + w_{end} \quad (11)$$

Where w_{now} is the inertia weight in $Iter$ th iteration. Then update the velocity of each particle as,

$$V_{iter} = w_{now} * V_{iter-1} + C_1 * R_1 * (pbest - X_{iter-1}) + C_2 * R_2 * (gbest - X_{iter-1}) \quad (12)$$

Where R_1 and R_2 are random values between $[0,1]$, $pbest$ is the ever achieved best position of this particle, X_{iter-1} is the position of last iteration. Afterward, the position of each particle is updated as,

$$X_{iter} = X_{iter-1} + V_{iter} \quad (13)$$

Finally, the best position of each particle $pbest$ and the global optimum of the swarm $gbest$ are updated.

Step 4. Judge the stopping criteria: If the number of iterations is larger than $MaxIter$, the iterations stop and $gbest$ is the final solution, else return to Step 3 and go on.

E. Feature Extraction and Classification

For each trial, the segmentations of EEG signals of C3 and C4 in optimal time interval are first obtained, then DTCWT is used to extract 8-16Hz signal component, finally three kinds of features are extracted, namely, Average Energy (AE), Root

Mean Square (RMS) and Signal Variance (SV). They are defined as below

$$E_{AE} = \frac{1}{m} \sum_{i=1}^m |x_i|^2 \quad (14)$$

$$E_{RMS} = \sqrt{\frac{x_1^2 + x_2^2 + \dots + x_m^2}{m}} \quad (15)$$

$$E_{SV} = \frac{1}{m} \sum_{i=1}^m (x_i - \bar{x})^2 \quad (16)$$

Where m is the number of sample points in the optimal time interval, x_i is the signal amplitude of i th sample point, $\bar{x}$ is the average signal amplitude. The features of EEG signal of C3 are denoted as $E_{AE} 3$, $E_{RMS} 3$ and $E_{SV} 3$. The features of EEG signal of C4 are denoted as $E_{AE} 4$, $E_{RMS} 4$ and $E_{SV} 4$. For each kind of feature, it is two dimensional with one component representing the feature of C3 and the other one representing the feature of C4.

LDA is used for classification in this study, the LDA approach introduced by Fisher aimed to find the optimal direction to project data upon and maximize Fisher's ratio. For samples of two classes $K_i (i=1,2)$ in the training set whose labels are known, the mean vector is calculated as

$$m_i = \frac{1}{N_i} \sum_{M \in K_i} M (i=1,2) \quad (17)$$

Where M is the feature vector, N_i is the number of samples of one class. Then, the between-class scatter matrix is obtained as

$$Z_i = \sum_{M \in K_i} (M - m_i)(M - m_i)^T (i=1,2) \quad (18)$$

$$Z_\omega = Z_1 + Z_2 \quad (19)$$

Afterward, the optimal vector ω^* and threshold ω_0 are calculated as

$$\omega^* = Z_\omega^{-1} (m_1 - m_2) \quad (20)$$

$$\omega_0 = (\omega^*)^T (m_1 + m_2) / 2 \quad (21)$$

Finally, the testing samples whose labels are unknown can be classified as in

$$y = \omega^* \cdot M - \omega_0 \quad (22)$$

III. EXPERIMENTAL RESULTS AND DISCUSSION

Based on the training set, DTCWT was used to extract sensory motor rhythms of EEG signal and PSO was used to search the optimal time interval for classification. Fig. 4 shows the fitness value evolution in PSO iteration procedure, note that AE feature is used here. From Fig. 4 we can see that, after particle initialization, the best cross-validation accuracy is below 82.4% and this value inclines to a constant after 5

epochs, which demonstrate that PSO has good convergence. Moreover, when RMS and SV are selected as features, the numbers of epochs for convergence are also less than 9 epochs. According to the best fitness value, the optimal time interval for classification can be obtained.

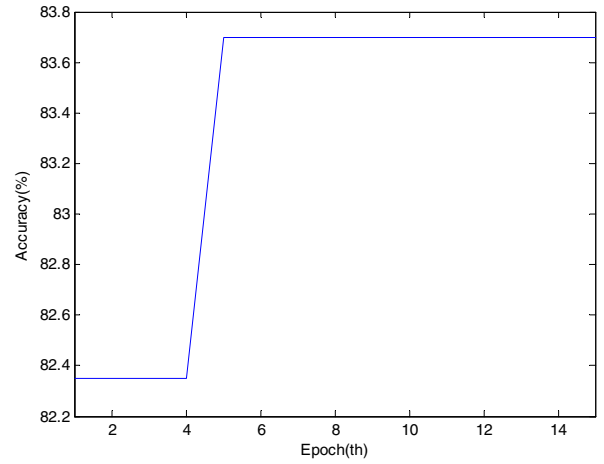


Fig. 4. Fitness value evolution in PSO

After the optimal time interval was determined, LDA classifier was obtained based on the training set. Fig. 5 shows the LDA classifier when AE is selected as feature, here the dotted line represents projection axis and the solid line means projection direction. It can be seen that, when these two-dimensional feature samples are projected onto the projection axis, samples of two classes can be discriminated well.

LDA classifier was then used to classify the testing samples, Table I presents the cross-validation accuracies on training set, the classification accuracies on testing set and obtained optimal time intervals when three different features are used. From Table I we can see that the relatively optimal time interval can be determined automatically by PSO and the classification accuracy reaches 90% when AE is selected as feature. In order to further justify the effectiveness of our approach, we also compared it with BCI competition winners(winner 89.21%, first runner-up 84.29%, second runner-up 82.86%). The improvement achieved by our method is 0.79% in comparison with the BCI competition winner.

TABLE I. PERFORMANCES OF DIFFERENT FEATURES

Feature	Accuracies on training set(%)	Accuracies on testing set(%)	Optimal time interval(point)
AE	83.7 ± 6.43	90	512-848
RMS	86.69 ± 6.21	86.43	512-768
SV	82.25 ± 6.73	87.14	528-816

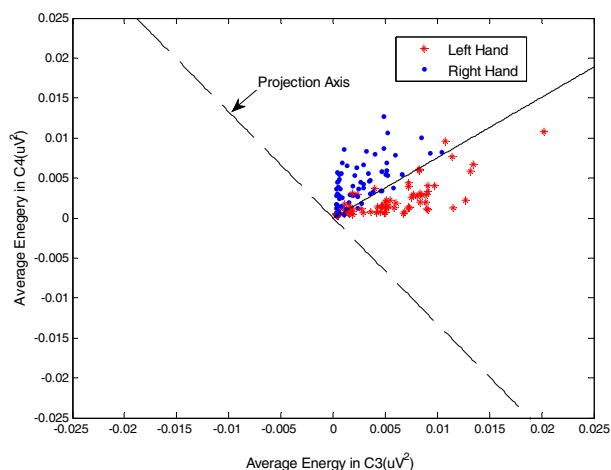


Fig. 5. LDA classifier

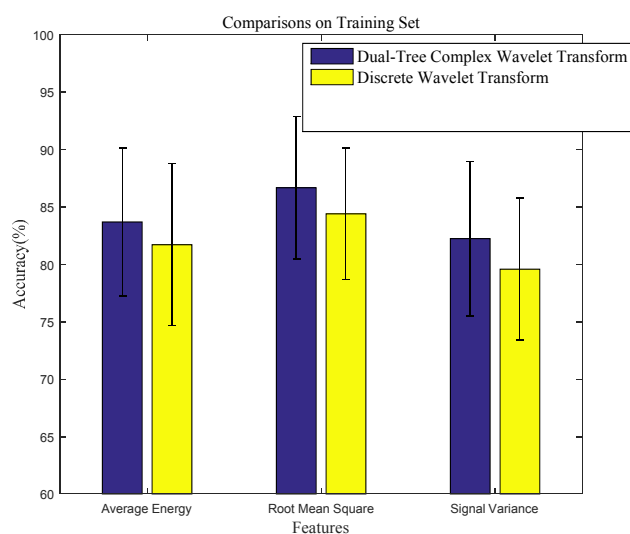


Fig. 6. Comparisons of DTCWT and DWT on training set

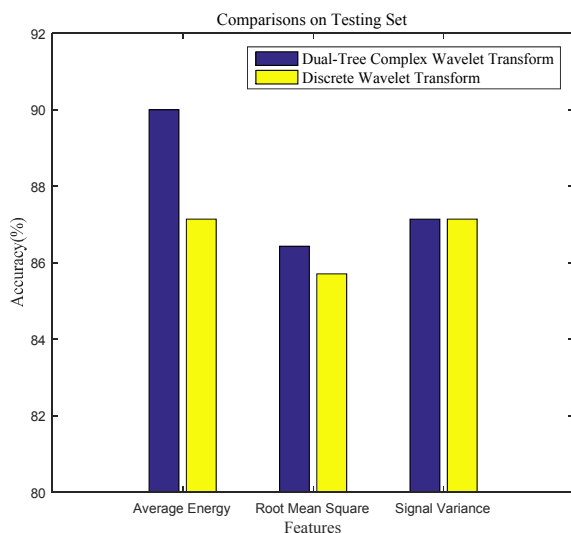


Fig. 7. Comparisons of DTCWT and DWT on testing set

To assess the superiority of DTCWT, we also used DWT to extract 8-16Hz signal component for further feature extraction and classification. Fig. 6 presents the comparisons of the classification performances on the training set. Fig. 7 presents the comparisons of classification performances on the testing set. It can be seen from these two figures that DTCWT performs better or at a same level than DWT. Which demonstrates that DTCWT with anti aliasing property is more suitable for sensory motor rhythms extraction.

IV. CONCLUSION

DWT is widely used in BCI systems due to its good performance in time-frequency analysis, however it has some inherent shortages such as shift variance and aliasing. Another important issue in BCI is how to select the most useful time interval for further processing. Aiming at these two problems, this paper propose a MI EEG pattern recognition approach based on DTCWT and PSO. DTCWT can avoid spectrum aliasing and leakage problems effectively and it is used for sensory motor rhythms extraction. PSO is applied to search optimal time interval for classification without any prior knowledge. The proposed method optimize features in both time and frequency domains and experimental results demonstrate its effectiveness.

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